

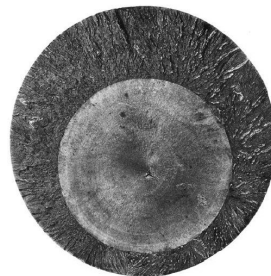
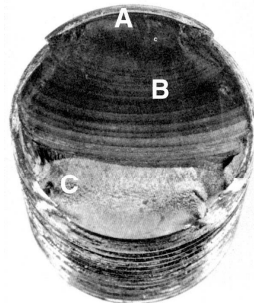
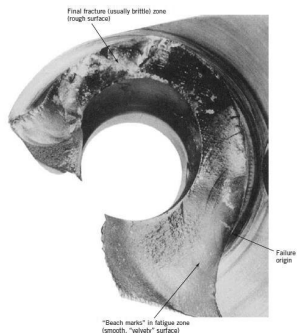
FATIGUE FAILURE

MCL742 DESIGN AND OPTIMIZATION
DEPARTMENT OF MECHANICAL ENGINEERING
IIT DELHI

Fatigue Failure

- Stresses vary with time or they fluctuate between different levels
- *variable, repeated, alternating, or fluctuating stresses*
- Machine members are found to have failed under the action of repeated or fluctuating stresses
- well below the ultimate strength of the material, and quite frequently even below the yield strength
- The most distinguishing characteristic of these failures is that the stresses have been repeated a very large number of times. Hence the failure is called a *fatigue failure*.
- Fatigue failure gives no warning! It is sudden and total, and hence dangerous.

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Fatigue Failure

- Stage I. Initiation of one or more microcracks
- Stage II. Progresses from microcracks to macrocracks forming parallel plateau-like fracture surfaces separated by longitudinal ridges - *beach marks* or *clamshell marks*
- Stage III. Sudden and fast fracture

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Fatigue Failure

Fatigue failure is due to *crack formation and propagation*. A fatigue crack will typically initiate at a discontinuity in the material where the cyclic stress is a maximum. Discontinuities can arise because of:

- Design of rapid changes in cross section, keyways, holes, etc. where stress concentrations occur.
- Elements that roll and/or slide against each other (bearings, gears, cams, etc.) under high contact pressure, developing concentrated subsurface contact stresses that can cause surface pitting or spalling after many cycles of the load.

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Fatigue Failure

- Carelessness in locations of stamp marks, tool marks, scratches, and burrs; poor joint design; improper assembly; and other fabrication faults.
- Composition of the material itself as processed by rolling, forging, casting, extrusion, drawing, heat treatment, etc. Microscopic and submicroscopic surface and subsurface discontinuities arise, such as inclusions of foreign material, alloy segregation, voids, hard precipitated particles, and crystal discontinuities.

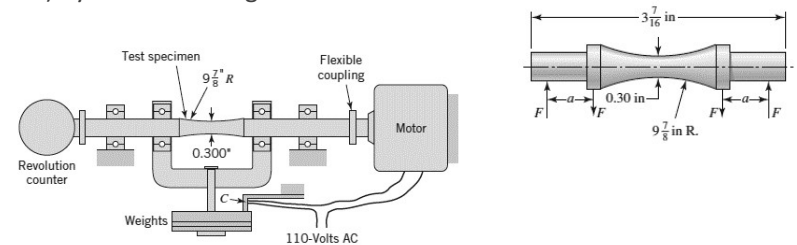
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Standard Fatigue Strength

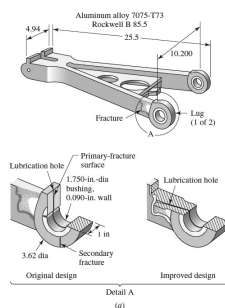
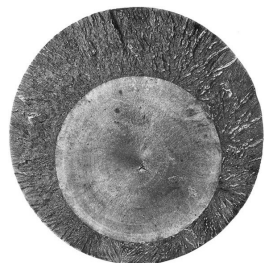
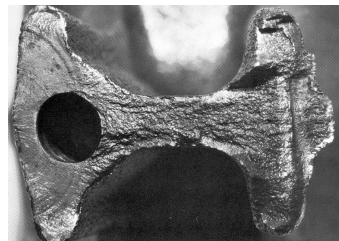
Specimens are subjected to *repeated or varying forces of specified magnitudes* while the cycles or stress reversals are counted to destruction.

R. R. Moore high-speed rotating-beam machine

The machine subjects the specimen to pure bending (no transverse shear) by means of weights.



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To establish the fatigue strength of a material, quite a number of tests are necessary because of the *statistical nature of fatigue*.

For the rotating-beam test, a constant bending load is applied, and the number of revolutions (*stress reversals*) of the beam required for failure is recorded.

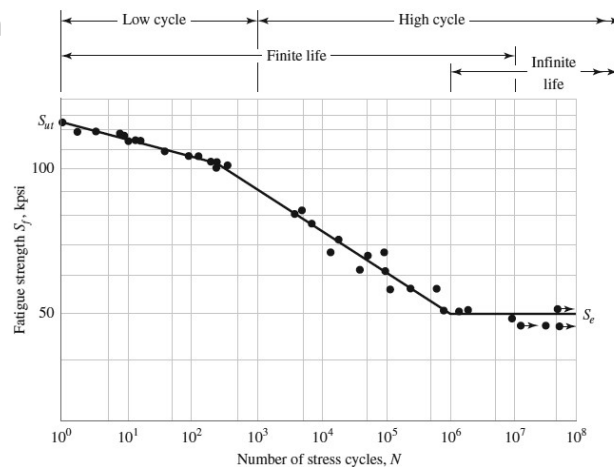
The first test is made at a stress that is somewhat under the ultimate strength of the material.

The second test is made at a stress that is less than that used in the first.

This process is continued, and the results are plotted as an *S-N* diagram.

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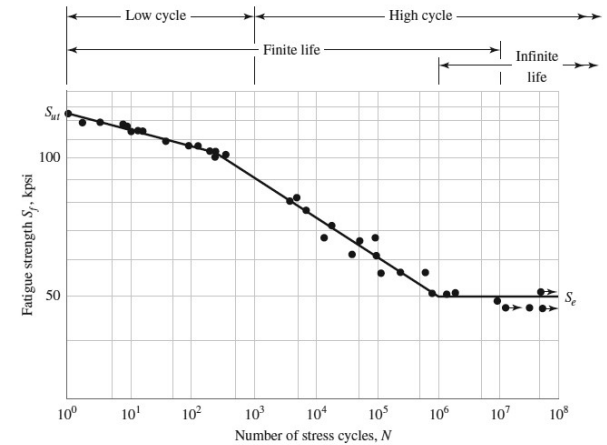
S-N diagram



An S-N diagram plotted from the results of completely reversed axial fatigue tests.
 Material: UNS G41300 steel, normalized;
 $S_{ut} = 116$ kpsi; maximum $S_{ut} = 125$ kpsi.

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The S-N diagram is usually obtained by completely reversed stress cycles, in which the stress level alternates between equal magnitudes of tension and compression.

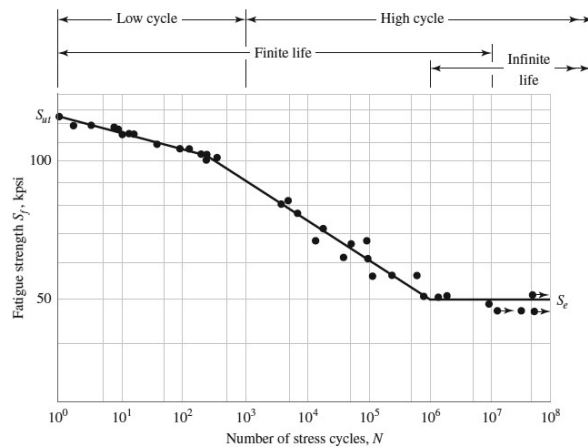


$N = 1$ to $N = 1000$ cycles is generally classified as *low-cycle fatigue*.
High-cycle fatigue is concerned with failure corresponding to stress cycles greater than 10^3 cycles.

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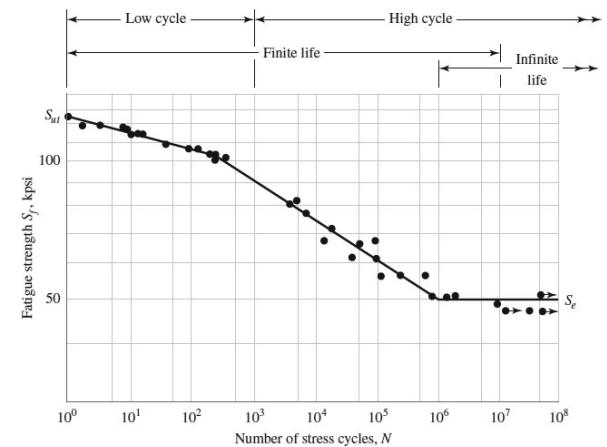
S-N diagram

The ordinate of the S-N diagram is called the *fatigue strength* S_f ; a statement of this strength value must *always be accompanied by a statement of the number of cycles* N to which it corresponds.



In the case of the steels, a knee occurs in the graph, and beyond this knee failure will not occur, no matter how great the number of cycles. The strength corresponding to the knee is called the *endurance limit* S_e , or the *fatigue limit*.

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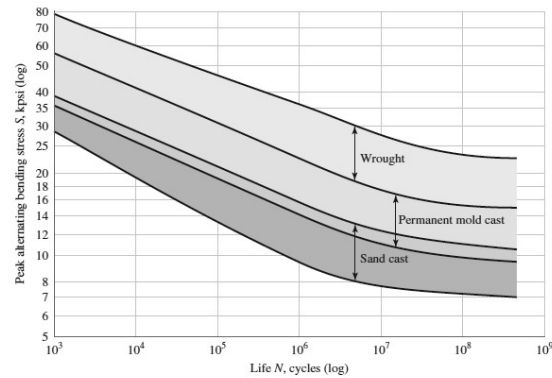
- *finite-life region*
- *infinite-life region*

- not clearly defined except for a specific material
- lies somewhere between 10^6 and 10^7 cycles for steels

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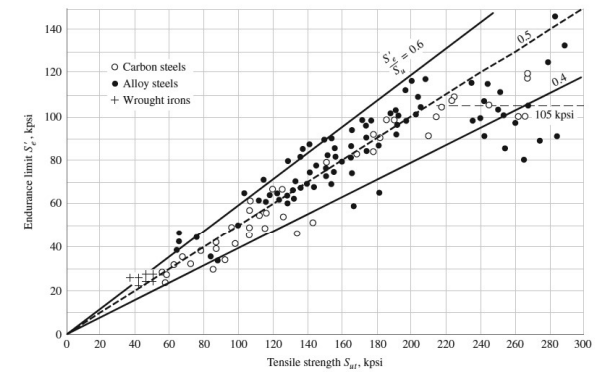
The graph never does become horizontal for nonferrous metals and alloys, and hence these materials do not have an endurance limit.

Since aluminum does not have an endurance limit, normally the fatigue strength S_f is reported at a specific number of cycles, normally $N = 5(10^8)$ cycles of reversed stress.



S - N bands for representative aluminum alloys, excluding wrought alloys with $S_u < 38$ kpsi.

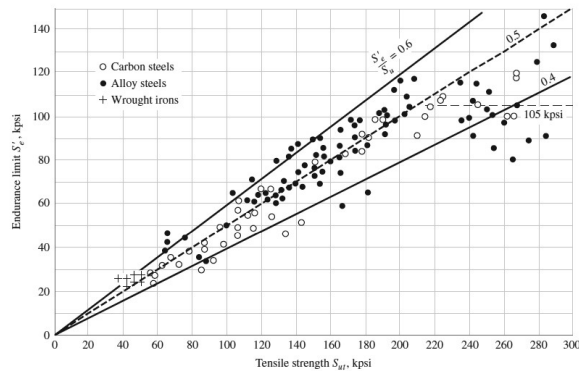
The Endurance Limit



For carbon and alloy steels,

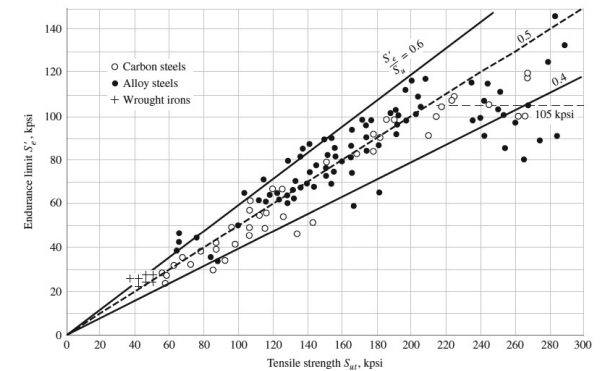
$$S'_e = \begin{cases} 0.5 S_{ut} & S_{ut} \leq 200 \text{ kpsi (1400 MPa)} \\ 100 \text{ kpsi} & S_{ut} > 200 \text{ kpsi} \\ 700 \text{ MPa} & S_{ut} > 1400 \text{ MPa} \end{cases}$$

The Endurance Limit



Graph of endurance limits versus tensile strengths from actual test results for a large number of wrought irons and steels. Ratios of S'_e/S_{ut} of 0.60, 0.50, and 0.40 are shown by the solid and dashed lines.

The Endurance Limit



For carbon and alloy steels,

$$S'_e = \begin{cases} 0.5 S_{ut} & S_{ut} \leq 1400 \text{ MPa} \\ 700 \text{ MPa} & S_{ut} \geq 1400 \text{ MPa} \end{cases}$$

Endurance Limit Modifying Factors

- *Material*: composition, basis of failure, variability
- *Manufacturing*: method, heat treatment, fretting corrosion, surface condition, stress concentration
- *Environment*: corrosion, temperature, stress state, relaxation times
- *Design*: size, shape, life, stress state, speed, fretting, galling

Surface Factor k_a

$$k_a = aS_{ut}^b$$

where S_{ut} is the minimum tensile strength and a and b are to be found

Surface Finish	Factor a		Exponent b
	S_{ut} kpsi	S_{ut} MPa	
Ground	1.34	1.58	-0.085
Machined or cold-drawn	2.70	4.51	-0.265
Hot-rolled	14.4	57.7	-0.718
As-forged	39.9	272.	-0.995

Endurance Limit Modifying Factors

$$S_e = k_a k_b k_c k_d k_e k_f S'_e$$

- where
- k_a = surface condition modification factor
 - k_b = size modification factor
 - k_c = load modification factor
 - k_d = temperature modification factor
 - k_e = reliability factor¹³
 - k_f = miscellaneous-effects modification factor
 - S'_e = rotary-beam test specimen endurance limit
 - S_e = endurance limit at the critical location of a machine part in the geometry and condition of use

Size Factor k_b

$$k_b = \begin{cases} (d/0.3)^{-0.107} = 0.879d^{-0.107} & 0.11 \leq d \leq 2 \text{ in} \\ 0.91d^{-0.157} & 2 < d \leq 10 \text{ in} \\ (d/7.62)^{-0.107} = 1.24d^{-0.107} & 2.79 \leq d \leq 51 \text{ mm} \\ 1.51d^{-0.157} & 51 < d \leq 254 \text{ mm} \end{cases}$$

For axial loading there is no size effect, so

$$k_b = 1$$

Equivalent diameter d_e

an *equivalent diameter d_e* obtained by equating the volume of material stressed *at and above 95 percent of the maximum stress* to the same volume in the rotating-beam specimen

For a rotating round section, the 95 percent stress area is the area in a ring having an outside diameter d and an inside diameter of $0.95d$. So, designating the 95 percent stress area $A_{0.95\sigma}$, we have

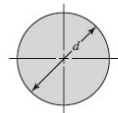
$$A_{0.95\sigma} = \frac{\pi}{4}[d^2 - (0.95d)^2] = 0.0766d^2$$

Loading Factor k_c

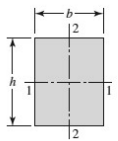
When fatigue tests are carried out with rotating bending, axial (push-pull), and torsional loading, the endurance limits differ with S_{ut}

$$k_c = \begin{cases} 1 & \text{bending} \\ 0.85 & \text{axial} \\ 0.59 & \text{torsion}^{17} \end{cases}$$

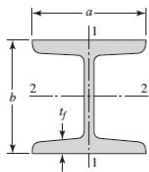
$A_{0.95\sigma}$ Areas of Common Nonrotating Structural Shapes



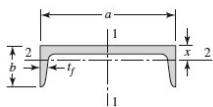
$$A_{0.95\sigma} = 0.01046d^2 \\ d_e = 0.370d$$



$$A_{0.95\sigma} = 0.05hb \\ d_e = 0.808\sqrt{hb}$$



$$A_{0.95\sigma} = \begin{cases} 0.10at_f & \text{axis 1-1} \\ 0.05ba & t_f > 0.025a \text{ axis 2-2} \end{cases}$$



$$A_{0.95\sigma} = \begin{cases} 0.05ab & \text{axis 1-1} \\ 0.052xa + 0.1t_f(b-x) & \text{axis 2-2} \end{cases}$$

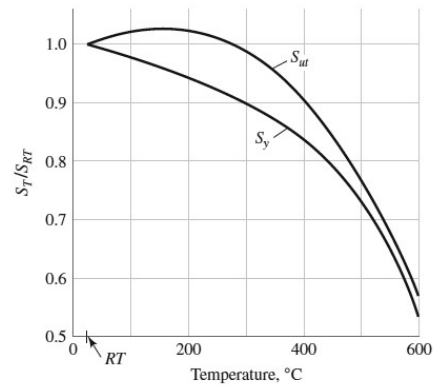
Temperature Factor k_d

When operating temperatures are below room temperature, brittle fracture is a strong possibility and should be investigated first.

When the operating temperatures are higher than room temperature, yielding should be investigated first because the yield strength drops off so rapidly with temperature.

Any stress will induce creep in a material operating at high temperatures; so this factor must be considered too.

A plot of the results of 145 tests of 21 carbon and alloy steels showing the effect of operating temperature on the yield strength S_y and the ultimate strength S_{ut} . The ordinate is the ratio of the strength at the operating temperature to the strength at room temperature.



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If the rotating beam endurance limit is known at room temperature, then use

$$k_d = \frac{S_T}{S_{RT}}$$

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Effect of Operating Temperature on the Tensile Strength of Steel.* (S_T = tensile strength at operating temperature; S_{RT} = tensile strength at room temperature; $0.099 \leq \hat{\sigma} \leq 0.110$)

Temperature, °C	S_T/S_{RT}	Temperature, °F	S_T/S_{RT}
20	1.000	70	1.000
50	1.010	100	1.008
100	1.020	200	1.020
150	1.025	300	1.024
200	1.020	400	1.018
250	1.000	500	0.995
300	0.975	600	0.963
350	0.943	700	0.927
400	0.900	800	0.872
450	0.843	900	0.797
500	0.768	1000	0.698
550	0.672	1100	0.567
600	0.549		

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Reliability Factor k_e

$$k_e = 1 - 0.08 z_a$$

Reliability Factors k_e Corresponding to 8 Percent Standard Deviation of the Endurance Limit

Reliability, %	Transformation Variate z_a	Reliability Factor k_e
50	0	1.000
90	1.288	0.897
95	1.645	0.868
99	2.326	0.814
99.9	3.091	0.753
99.99	3.719	0.702
99.999	4.265	0.659
99.9999	4.753	0.620

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Miscellaneous-Effects Factor k_f

Residual stresses

Case hardened

Corrosion

Electrolytic plating

Metal spraying

Cyclic frequency

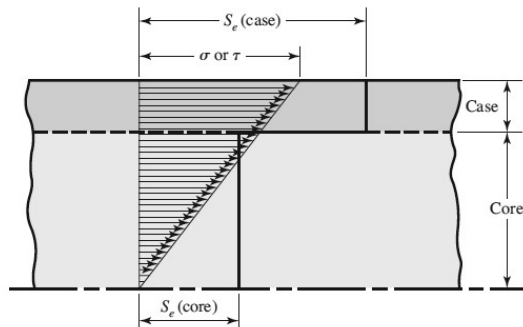
Fretting corrosion

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Stress Concentration and Notch Sensitivity

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The failure of a case-hardened part in bending or torsion. In this example, failure occurs in the core.



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Stress Concentration

Any discontinuity in a machine part alters the stress distribution in the neighborhood of the discontinuity.

Such discontinuities are called stress raisers, and the regions in which they occur are called areas of stress concentration.

Stress concentrations can also arise from some irregularity not inherent in the member, such as tool marks, holes, notches, grooves, or threads.

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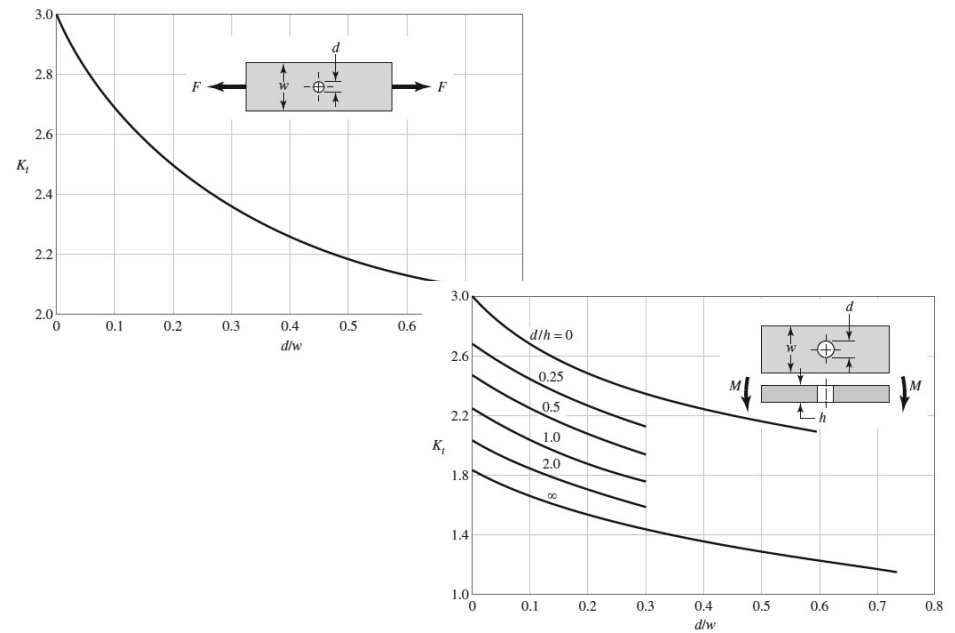
Stress Concentration

A *theoretical, or geometric, stress-concentration factor* K_t or K_{ts} is used to relate the actual maximum stress at the discontinuity to the *nominal stress*.

The factors are defined by the equations

$$K_t = \frac{\sigma_{\max}}{\sigma_0} \quad K_{ts} = \frac{\tau_{\max}}{\tau_0}$$

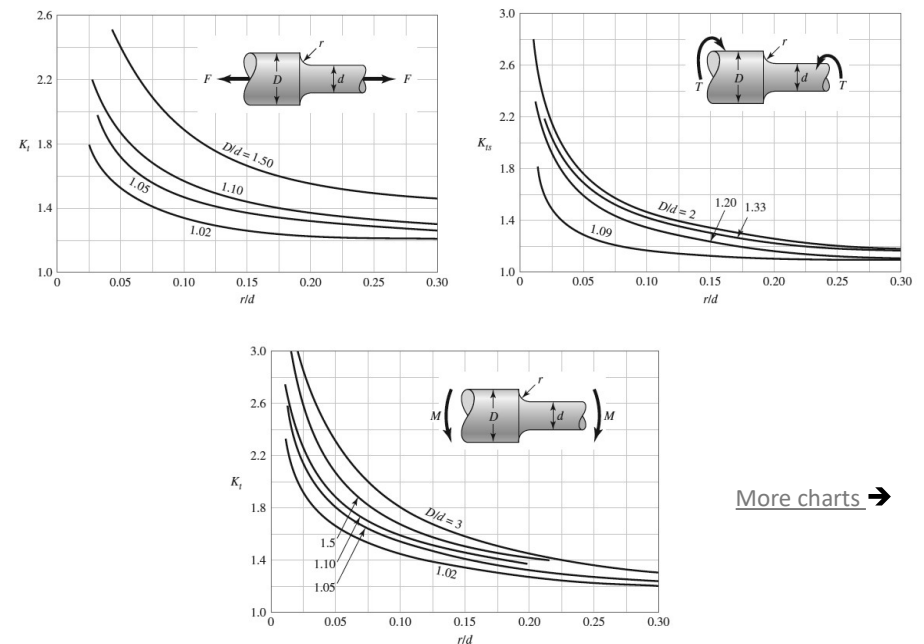
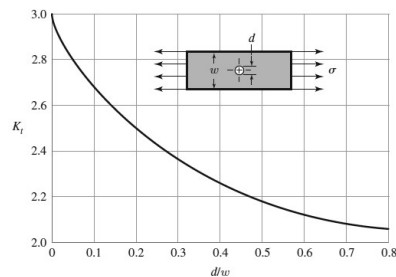
where K_t is used for normal stresses and K_{ts} for shear stresses. The nominal stress σ_0 or τ_0 is the stress calculated by using the elementary stress equations and the net area, or net cross section.



Stress Concentration

The stress-concentration factor depends for its value only on the geometry of the part. That is, the particular material used has no effect on the value of K_t . This is why it is called a theoretical stress-concentration factor.

Most stress-concentration factors are found by using experimental techniques.



[More charts](#) →

Stress Concentration

In *static loading*, stress-concentration factors are applied as follows.

In *ductile materials* ($\varepsilon_f \geq 0.05$), the stress-concentration factor *is not* usually applied to predict the critical stress, because plastic strain in the region of the stress is localized and has a strengthening effect.

In *brittle materials* ($\varepsilon_f < 0.05$), the geometric stress-concentration factor K_t is applied to the nominal stress before comparing it with strength.

For *dynamic loading*, the stress concentration effect is significant for *both* ductile and brittle materials and must always be taken into account.

Notch Sensitivity

Notch sensitivity q is defined by the equation

$$q = \frac{K_f - 1}{K_t - 1} \quad \text{or} \quad q_{\text{shear}} = \frac{K_{fs} - 1}{K_{ts} - 1}$$

where q is usually between zero and unity.

If $q = 0$, then $K_f = 1$, and the material has no sensitivity to notches at all.

On the other hand, if $q = 1$, then $K_f = K_t$, and the material has full notch sensitivity.

Notch Sensitivity

Notch sensitivity q is defined by the equation

$$q = \frac{K_f - 1}{K_t - 1} \quad \text{or} \quad q_{\text{shear}} = \frac{K_{fs} - 1}{K_{ts} - 1}$$

where K_t (or K_{ts}) is the theoretical stress-concentration factor, and K_f is commonly called a *fatigue stress-concentration factor*

$$K_f = \frac{\text{maximum stress in notched specimen}}{\text{stress in notch-free specimen}}$$

Notch Sensitivity

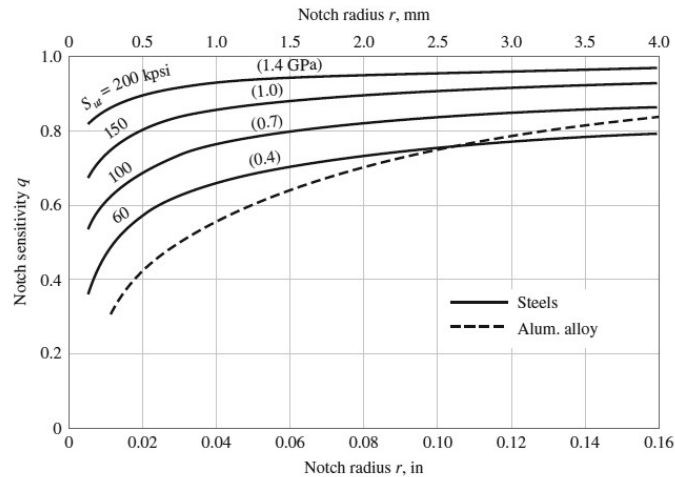
In analysis or design work,

→ Find K_t first, from the geometry of the part.

→ Then specify the material, find q , and solve for K_f from the equation

$$K_f = 1 + q(K_t - 1) \quad \text{or} \quad K_{fs} = 1 + q_{\text{shear}}(K_{ts} - 1)$$

Notch-sensitivity charts for steels and UNS A92024-T wrought aluminum alloys subjected to reversed bending or reversed axial loads.



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Notch Sensitivity

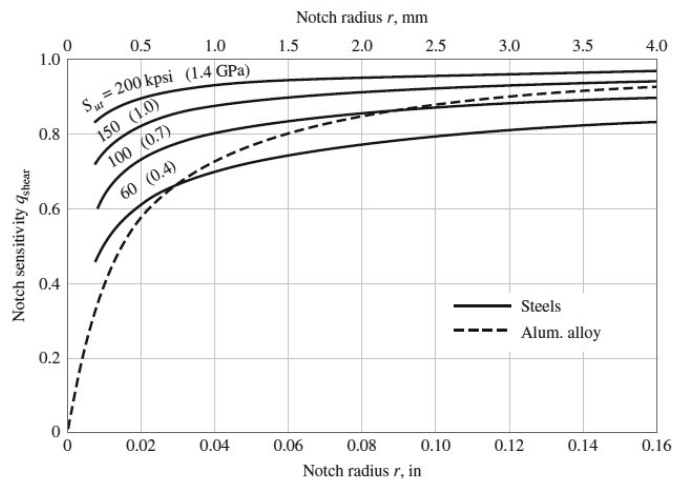
The effective maximum stress in fatigue is

$$\sigma_{\max} = K_f \sigma_0 \quad \text{or} \quad \tau_{\max} = K_{fs} \tau_0$$

where K_f is a reduced value of K_t and σ_0 is the nominal stress.

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Notch-sensitivity curves for materials in reversed torsion.



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Notch Sensitivity

Neuber equation

$$K_f = 1 + \frac{K_t - 1}{1 + \sqrt{a/r}}$$

where $\sqrt{a}$ is defined as the *Neuber constant* and is a material constant.

The notch sensitivity equation is

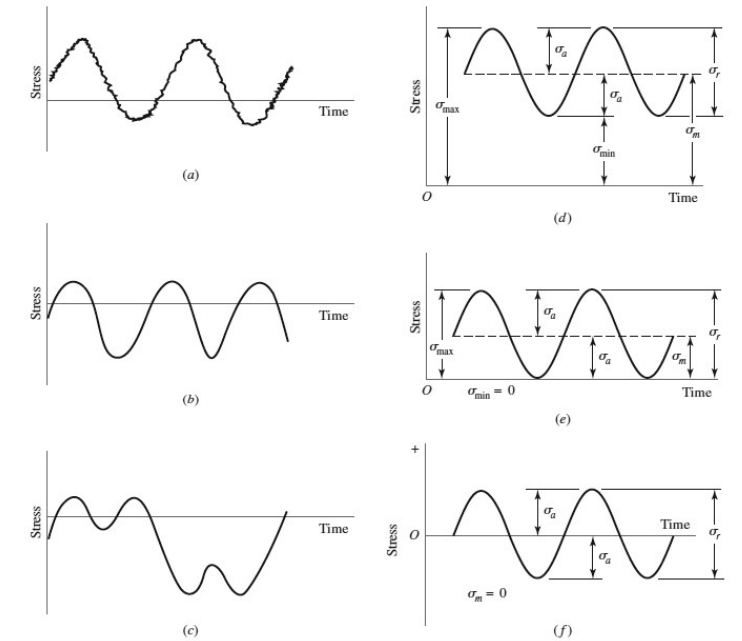
$$q = \frac{1}{1 + \frac{\sqrt{a}}{\sqrt{r}}}$$

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Characterizing Fluctuating Stresses

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Some stress-time relations:
 (a) fluctuating stress with high frequency ripple;
 (b and c) non-sinusoidal fluctuating stress;
 (d) sinusoidal fluctuating stress;
 (e) repeated stress;
 (f) completely reversed sinusoidal stress.



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If the largest force is F_{max} and the smallest force is F_{min} , then a steady component and an alternating component can be constructed as follows:

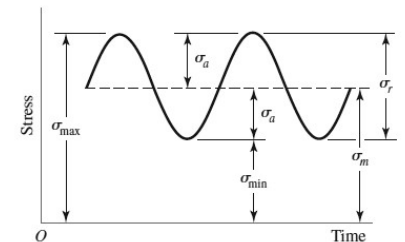
$$F_m = \frac{F_{max} + F_{min}}{2} \quad F_a = \left| \frac{F_{max} - F_{min}}{2} \right|$$

where F_m is the midrange steady component of force, and F_a is the amplitude of the alternating component of force.

σ_{min} = minimum stress σ_m = midrange component
 σ_{max} = maximum stress σ_r = range of stress
 σ_a = amplitude component σ_s = static or steady stress

$$\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2}$$

$$\sigma_a = \left| \frac{\sigma_{max} - \sigma_{min}}{2} \right|$$



The steady, or static, stress is not the same as the midrange stress; in fact, it may have any value between σ_{min} and σ_{max} .

The steady stress exists because of a fixed load or preload applied to the part, and it is usually independent of the varying portion of the load.

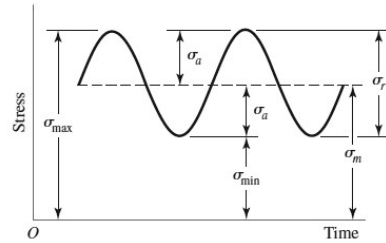
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σ_{min} = minimum stress σ_m = midrange component
 σ_{max} = maximum stress σ_r = range of stress
 σ_a = amplitude component σ_s = static or steady stress

$$\sigma_m = \frac{\sigma_{max} + \sigma_{min}}{2}$$

$$\sigma_a = \left| \frac{\sigma_{max} - \sigma_{min}}{2} \right|$$

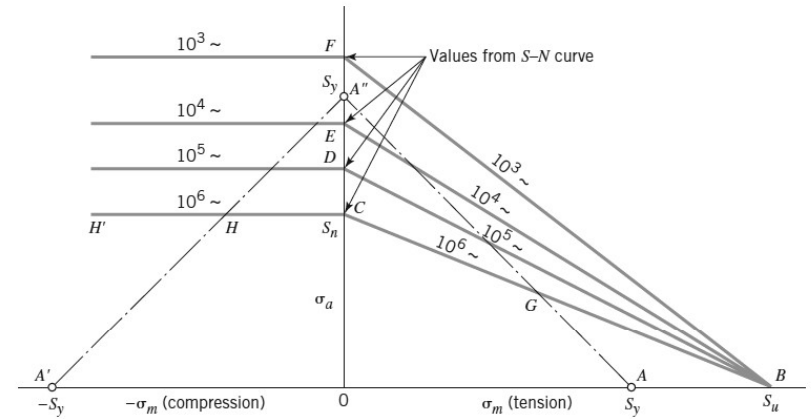


stress ratio

$$R = \frac{\sigma_{min}}{\sigma_{max}}$$

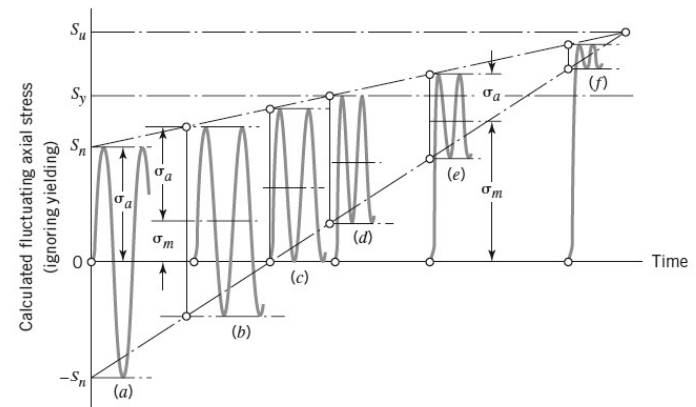
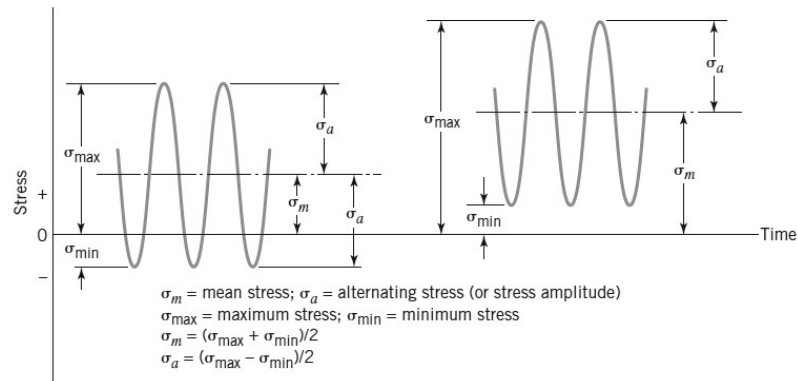
amplitude ratio

$$A = \frac{\sigma_a}{\sigma_m}$$

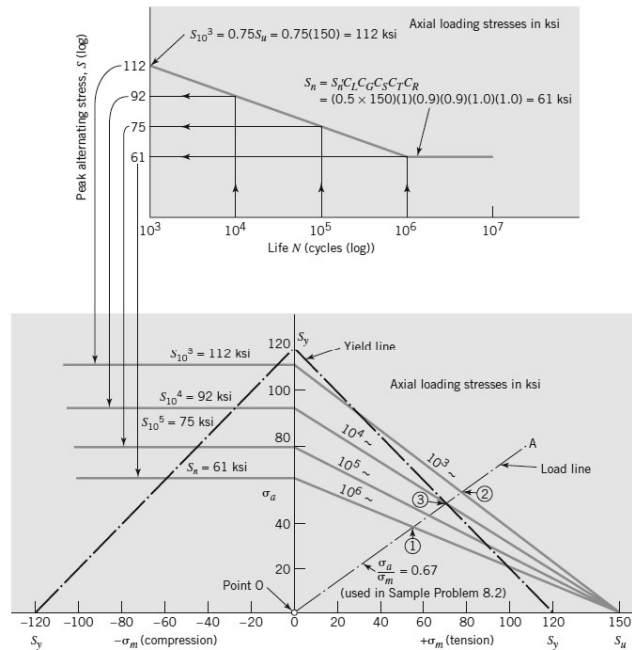


Constant-life fatigue diagram - ductile materials

Effect of Mean Stress on Fatigue Strength

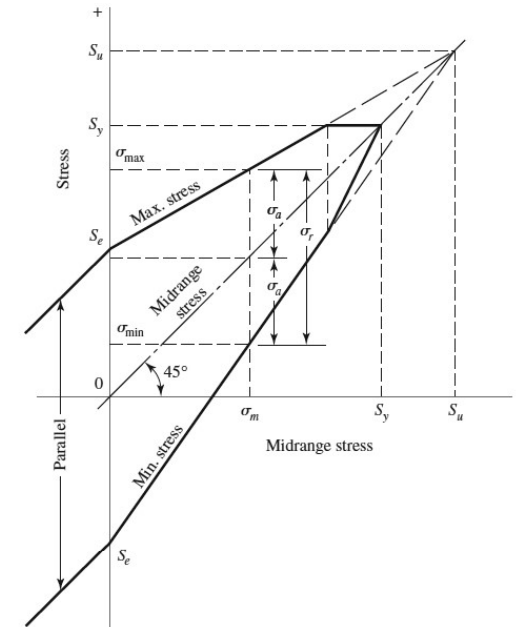


Various fluctuating uniaxial stresses, all of which correspond to equal fatigue life



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Modified Goodman diagram showing all the strengths and the limiting values of all the stress components for a particular midrange stress.

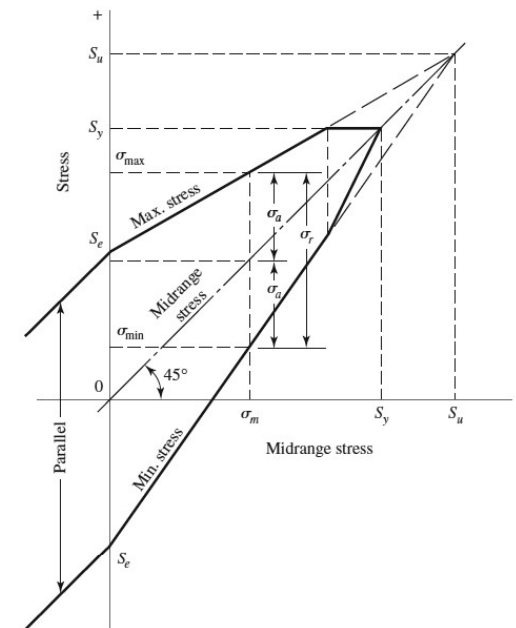


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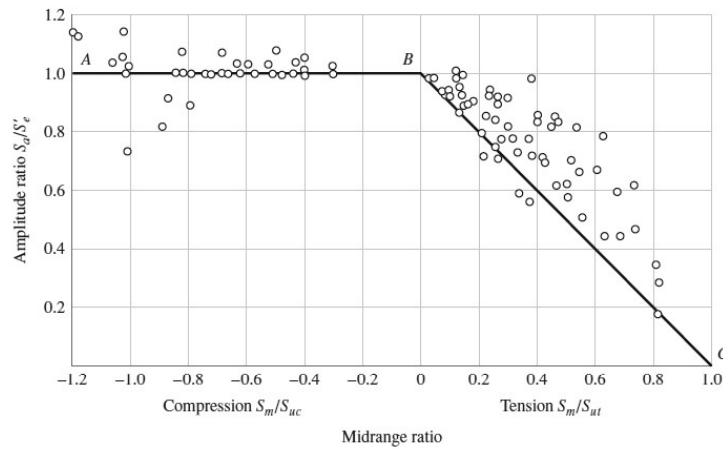
Fatigue Failure Criteria for Fluctuating Stress

The modified Goodman diagram consists of the lines constructed to S_e (or S_f) above and below the origin.

Note that the yield strength is also plotted on both axes, because yielding would be the criterion of failure if σ_{max} exceeded S_y .

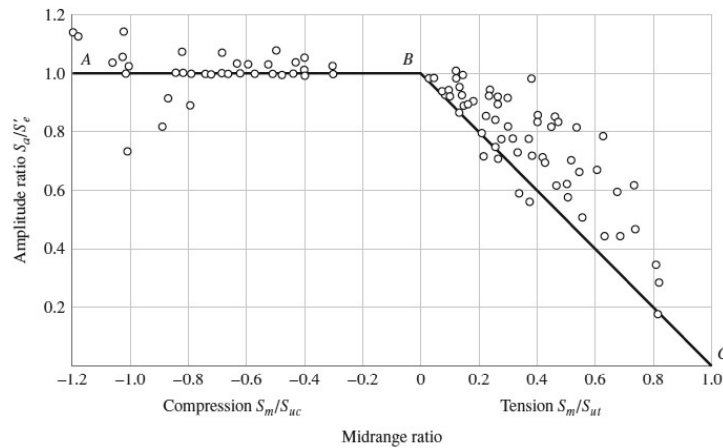
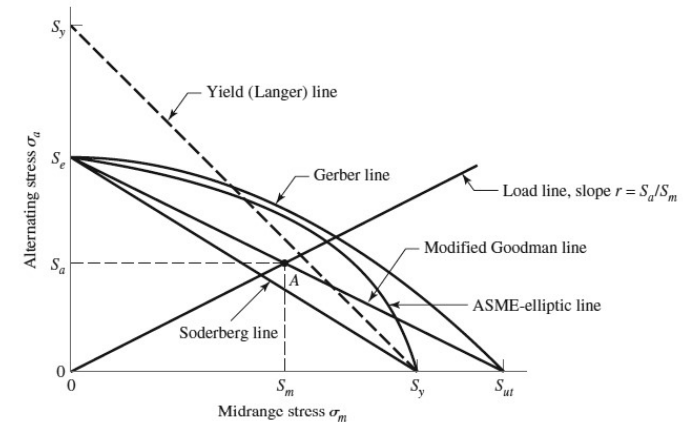


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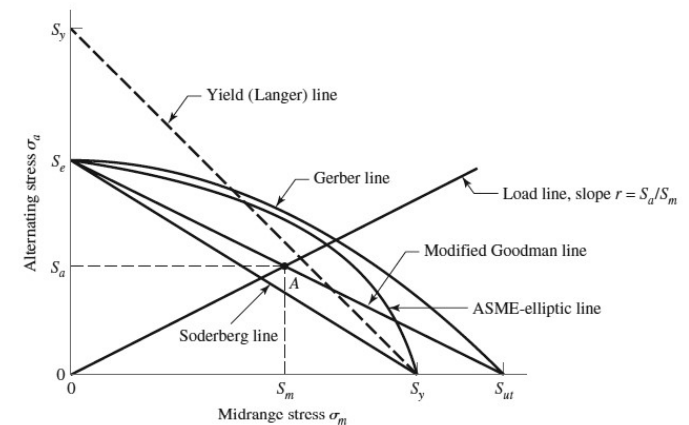
Plot of fatigue failures for midrange stresses in both tensile and compressive regions. Normalizing the data by using the ratio of steady strength component to tensile strength S_m/S_{ut} , steady strength component to compressive strength S_m/S_{uc} and strength amplitude component to endurance limit S_a/S_e' enables a plot of experimental results for a variety of steels.

Fatigue diagram showing various criteria of failure. For each criterion, points on or “above” the respective line indicate failure. Some point A on the Goodman line, for example, gives the strength S_m as the limiting value of σ_m corresponding to the strength S_a , which, paired with σ_m , is the limiting value of σ_a .



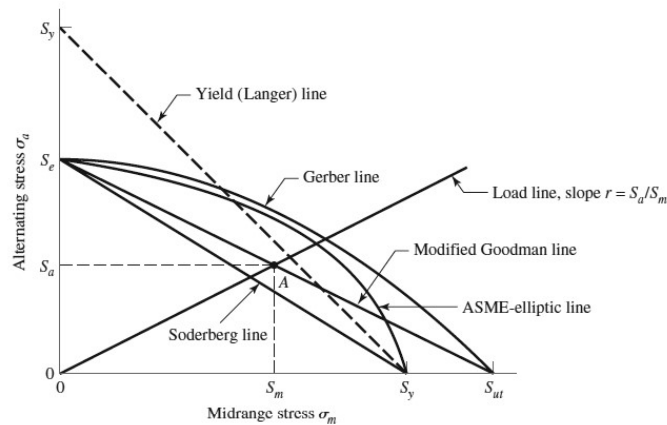
Here the abscissa represents the ratio of the midrange strength S_m to the ultimate strength, with tension plotted to the right and compression to the left. The ordinate is the ratio of the alternating strength to the endurance limit. The line BC then represents the *modified Goodman criterion of failure*. Note that the existence of midrange stress in the compressive region has little effect on the endurance limit.

Five criteria of failure:
the Soderberg, the modified Goodman,
the Gerber, the ASME-elliptic, and yielding

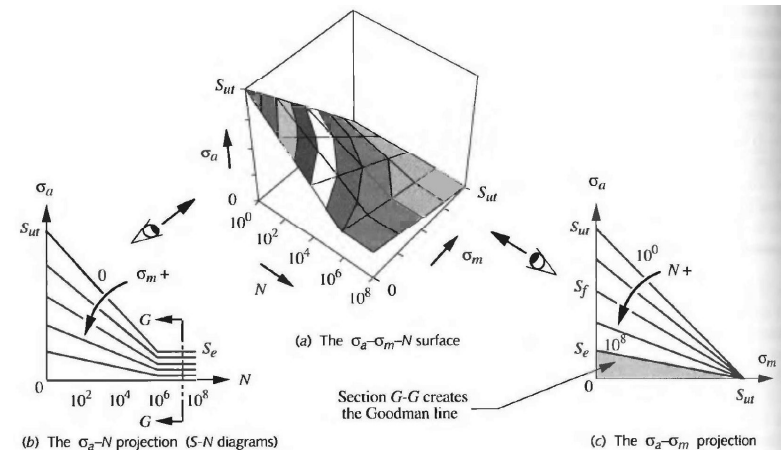


Soderberg line

$$\frac{S_a}{S_e} + \frac{S_m}{S_y} = 1 \quad \Rightarrow \quad \frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_y} = \frac{1}{n}$$



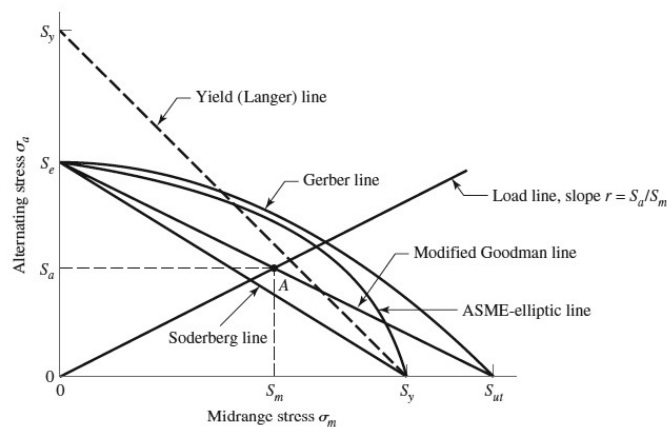
Creating the Modified-Goodman diagram



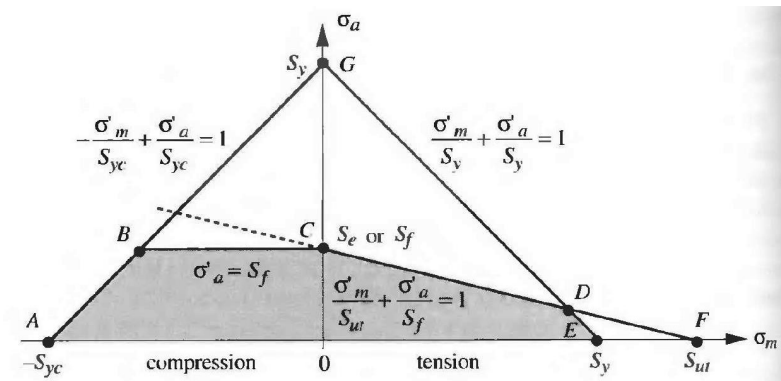
Effect of combination of mean and alternating stresses

modified Goodman

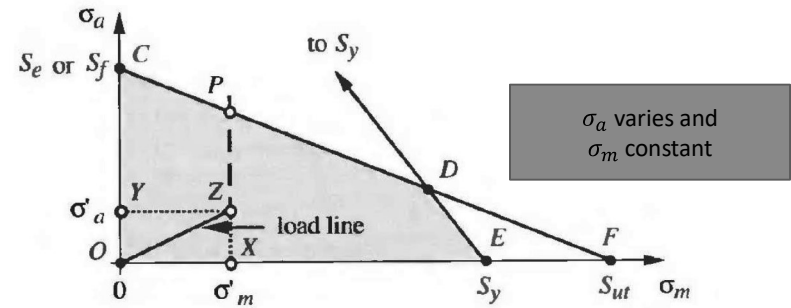
$$\frac{S_a}{S_e} + \frac{S_m}{S_{ut}} = 1 \quad \Rightarrow \quad \frac{\sigma_a}{S_e} + \frac{\sigma_m}{S_y} = \frac{1}{n}$$



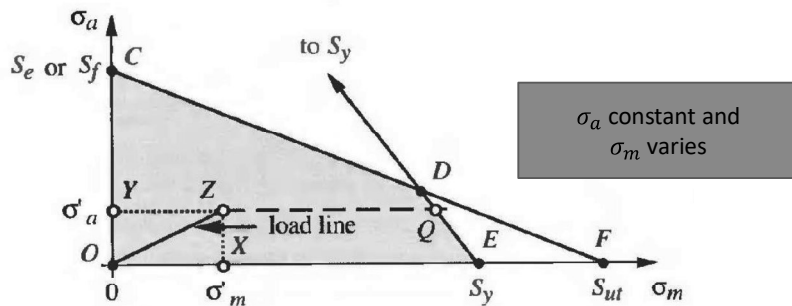
Augmented Modified-Goodman diagram



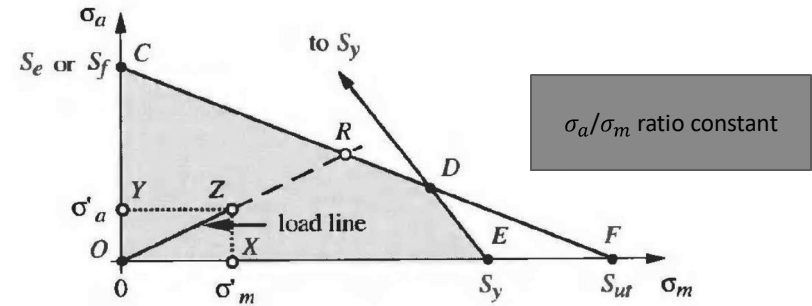
Determine the Safety Factor with Fluctuating Stresses



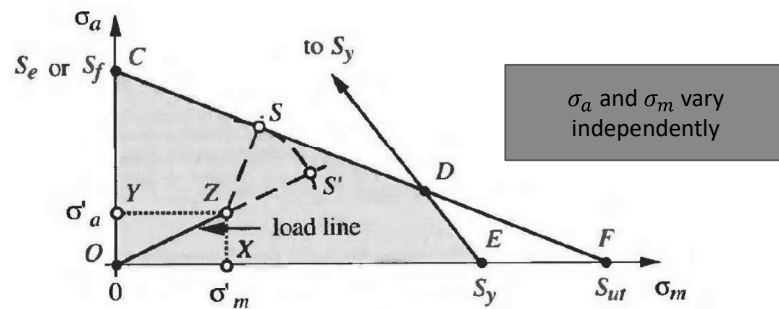
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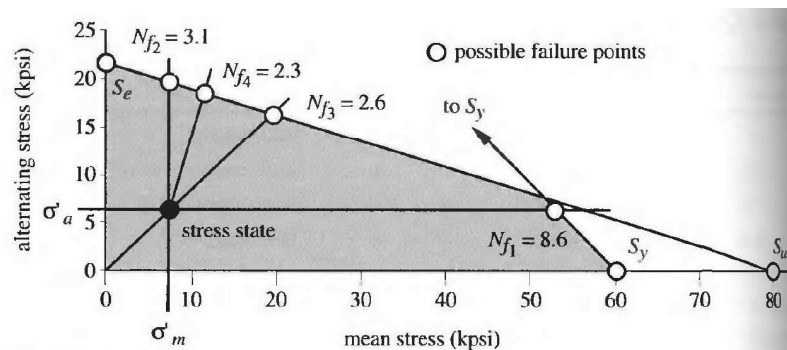


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Combinations of Loading Modes

Combinations of different types of loading, such as combined bending, torsion, and axial.



- The first step is to generate two stress elements—one for the alternating stresses and one for the midrange stresses.
- Apply the appropriate fatigue stress-concentration factors to each of the stresses; i.e., apply $(K_f)_{bending}$ for the bending stresses, $(K_{fs})_{torsion}$ for the torsional stresses, and $(K_f)_{axial}$ for the axial stresses.
- Next, calculate an equivalent von Mises stress for each of these two stress elements, σ'_a and σ'_m .
- Finally, select a fatigue failure criterion (modified Goodman, Gerber, ASME-elliptic, or Soderberg) to complete the fatigue analysis.

Considering that the bending, torsional, and axial stresses have alternating and midrange components, the von Mises stresses for the two stress elements can be written as

$$\sigma'_a = \left\{ \left[(K_f)_{\text{bending}} (\sigma_a)_{\text{bending}} + (K_f)_{\text{axial}} \frac{(\sigma_a)_{\text{axial}}}{0.85} \right]^2 + 3 \left[(K_{fs})_{\text{torsion}} (\tau_a)_{\text{torsion}} \right]^2 \right\}^{1/2}$$

$$\sigma'_m = \left\{ \left[(K_f)_{\text{bending}} (\sigma_m)_{\text{bending}} + (K_f)_{\text{axial}} (\sigma_m)_{\text{axial}} \right]^2 + 3 \left[(K_{fs})_{\text{torsion}} (\tau_m)_{\text{torsion}} \right]^2 \right\}^{1/2}$$

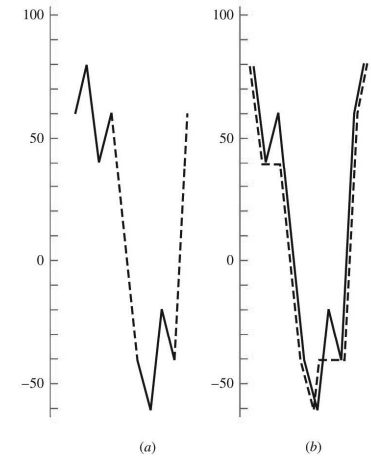
$$\sigma' = (\sigma_x^2 + 3\tau_{xy}^2)^{1/2}$$

Varying Fluctuating Stresses

Loading patterns may be complex

Simplifications may be necessary

Small fluctuations may be negligible compared to large cycles



Varying, Fluctuating Stresses; Cumulative Fatigue Damage

Cumulative Fatigue Damage

A common situation is to load at σ_1 for n_1 cycles, then at σ_2 for n_2 cycles, etc.

The cycles at each stress level contributes to the fatigue damage

Accumulation of damage is represented by the *Palmgren-Miner cycle-ratio summation rule*, also known as *Miner's rule*

$$\sum \frac{n_i}{N_i} = c$$

where n_i is the number of cycles at stress level σ_i and N_i is the number of cycles to failure at stress level σ_i

c is experimentally found to be in the range $0.7 < c < 2.2$, with an average value near unity

Cumulative Fatigue Damage

Defining D as the accumulated damage,

$$D = \sum \frac{n_i}{N_i}$$

When $D = c = 1$, failure ensues.

Use of the Miner rule to predict the endurance limit of a material that has been overstressed for a finite number of cycles.

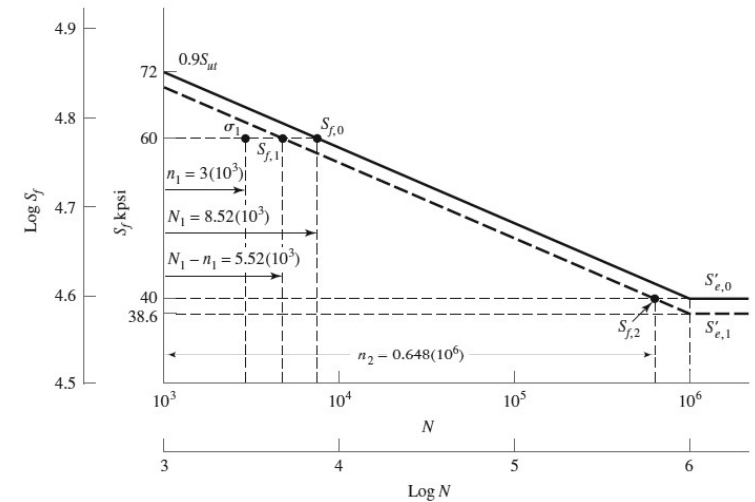
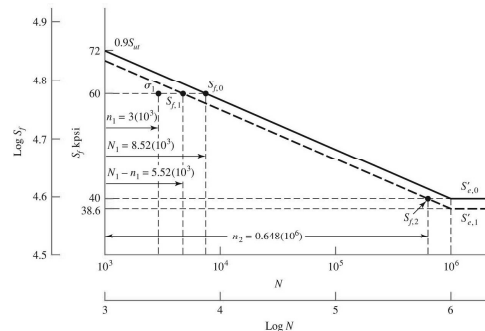


Illustration of Miner's Rule

Figure illustrates the effect of Miner's rule on endurance limit and fatigue failure line.

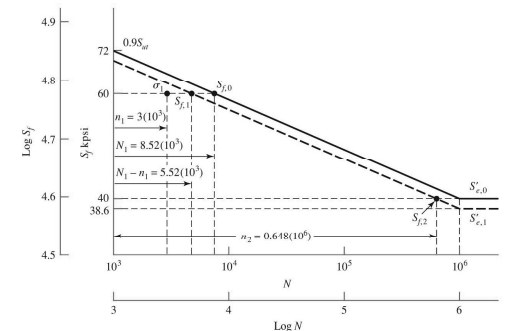
Note that the damaged material line is predicted to be parallel to original material line.



Weaknesses of Miner's Rule

Miner's rule fails to agree with experimental results in two ways

- It predicts the static strength S_{ut} is damaged.
- It does not account for the order in which the stresses are applied

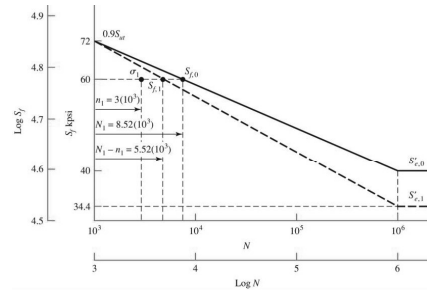


Manson's Method

Manson's method overcomes deficiencies of Miner's rule.

It assumes all fatigue lines on the S - N diagram converge to a common point at $0.9S_{ut}$ at 10^3 cycles.

It requires each line to be constructed in the same historical order in which the stresses occur.



REFERENCES

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Norton R.L., *Machine Design: An integrated approach*, 5th Edition, Pearson Education, 2018.

Juvinall & Marshek, *Fundamentals of Machine Component Design*, Fifth Edition, John Wiley & Sons, Inc., 2016.

Use of the Manson method to predict the endurance limit of a material that has been overstressed for a finite number of cycles.

